

JEE MAIN REVISION MASTER: SURFACE CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Surface Chemistry — Core Principles & Governing Laws	14 PYQs	Core Principle
Moderate	Surface Chemistry — Applied Cases, Reactions & Traps	5 PYQs	Applied Topics
Standard	Surface Chemistry — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Surface Chemistry — Core Principles & Governing Laws	$y=mx+c$, $\text{NaCl} > \text{BaCl}_2 < \text{AlCl}_3$, $\text{Na}_2\text{S} \rightarrow$
Surface Chemistry — Applied Cases, Reactions & Traps	$-\text{SO}_3^-$, I^- , 0.1M AgNO_3
Surface Chemistry — Standard Calculations & Problem Archetypes	$1.5 \times 10^{-4} \text{mol L}^{-1}$, mol L^{-1}

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Surface Chemistry — Core Principles & Governing Laws (High Yield — 14 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Surface Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Surface Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Surface Chemistry PYQs.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "For the coagulations of positively charged hydrated ferric - oxide sol, the flocculating power of the ions is in the order:..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Gold sol is not a..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "For Freundlich adsorption isotherm, a graph of $\log(x/m)$ vs $\log(p)$ gives a straight line. The slope of line and its Y -axis intercept r..." [Ans: N/A]

3.2 Surface Chemistry — Applied Cases, Reactions & Traps (Moderate Yield — 5 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Surface Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Identify the incorrect statement..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "A sol of AgI is prepared by mixing equal volumes of $0.1M AgNO_3$ and $0.2MKI$, which of the following statement is correct?..." [Ans: N/A]
- [2017 Shift] Q#1 (MCQ): "Which of the following is not a favourable condition for physical adsorption?..." [Ans: N/A]

3.3 Surface Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2019 Shift] Q#1 (MCQ): "Critical Micelle concentration for a soap solution is $1.5 \times 10^{-4} mol L^{-1}$. Micelle formation is possible only when the concentrat..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Identify the incorrect statement

- (A) Values of colligative properties of colloidal solution are of small order compared to values of true solution.
- (B) Tyndall effect is observed only when diameter of the dispersed particles is not much smaller than wavelength of incident light.
- (C) Colour of colloidal solution depends on the wavelength of light scattered by the dispersed particles.
- (D) Brownian movement is due to balanced bombardment of molecules of dispersion medium on colloidal particles.

Correct Answer: None

Solution Analysis: Among the given statements incorrect statement is given in option (d). Its correct form is "Brownian movement is due to unbalanced bombardment of molecules of dispersion medium on colloidal particles".

Benchmark Problem 2: KCET 2024 (Q#2)

For the coagulations of positively charged hydrated ferric - oxide sol, the flocculating power of the ions is in the order:

- (A) $\text{PO}_4^{3-} \rightarrow \text{SO}_4^{2-} \rightarrow \text{Cl}^- \rightarrow [\text{Fe}(\text{CN})_6]^{4-}$
- (B) $\text{Cl}^- \rightarrow \text{SO}_4^{2-} \rightarrow \text{PO}_4^{3-} \rightarrow [\text{Fe}(\text{CN})_6]^{4-}$
- (C) $\text{SO}_4^{2-} = \text{Cl}^- = \text{PO}_4^{3-} = [\text{Fe}(\text{CN})_6]^{4-}$
- (D) $[\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{PO}_4^{3-} \rightarrow \text{SO}_4^{2-} \rightarrow \text{Cl}^-$

Correct Answer: None

Solution Analysis: Generally, greater the valence of the flocculating ion added, the greater is its power to cause precipitation. This is called as Hardy-Schulze rule. So according to Hardy-Schulze rule the flocculating power of the ions is in the order $[\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{PO}_4^{3-} \rightarrow \text{SO}_4^{2-} \rightarrow \text{Cl}^-$.

Benchmark Problem 3: KCET 2019 (Q#1)

Critical Micelle concentration for a soap solution is $1.5 \times 10^{-4} \text{ mol L}^{-1}$. Micelle formation is possible only when the concentration of soap solution in mol L^{-1} is

- (A) 2.0×10^{-3}
- (B) 4.6×10^{-5}
- (C) 7.5×10^{-5}
- (D) 1.1×10^{-4}

Correct Answer: None

Solution Analysis: The formation of micelles takes place only above a particular concentration called critical micelle concentration (CMC). 2×10^{-3} concentration is higher than CMC value (i.e. $1.5 \times 10^{-4} \text{ mol/L}$). All other options have lower value of CMC. Hence, micelle formation is possible only at this concentration.